

Is electricity the Achilles heel of AI?

Posted by u/MrDinglehut • 0 points • 30 comments

I guess you could figure out like the number of chips that are built by Nvidia every year or how many data centers that are or will be built for AI and make a guess at the amount of electricity that would take and like do we even have enough electrical capacity to make it all run?

It would be nice to know how much everyone's electricity bill is going to go up ahead of time.

This sounds like the whole Achilles heel of the whole thing.

Comments

u/attrezzarturo • 2 points • 2025-09-30 15:35 UTC

nope

u/Sad-Celebration-7542 • 1 points • 2025-09-27 20:23 UTC

I don't think electricity matters much for AI. I imagine a few paths.

Path 1: AI becomes god. It's a trillion \$ industry. On this path, you don't give a shit what electricity costs. On this path, AI actually helps with the supply problem.

Path 2: AI is bullshit. It's a bubble that tanks some large tech companies. On this path, electricity is irrelevant because not much compute is built and what is built can be controlled dynamically since it's not actually important to have your meme RIGHT NOW. It becomes the metaverse and we move on.

Path 3: AI is medium useful. It's like email. We build data centers like we did before. Demand slowly grows.

u/PensiveinNJ • 2 points • 2025-09-27 17:34 UTC

The Achilles heel of AI is that it's shit at almost anything useful.

We are never going to as a society get anywhere until people internalize this.

It's failing because it sucks. It's failing because even giving it away isn't useful for many businesses.

Please internalize this.

u/griff1 • 1 points • 2025-09-27 14:55 UTC

Personally, I think materials are a bigger obstacle than people expect. Mostly because that's my field, but also because of just how pernicious the problem is. It feels like everyone is worried because they have some bottleneck, going down to stuff as fundamental as clothes. And the problem isn't simply the fragility of supply chains for the stuff needed to make GPUs or just keep life going, it's that all the money being spent on AI is consuming resources needed elsewhere. It's like a weed growing so much that it starves itself to death as well as kills off everything else.

u/jlks1959 • -1 points • 2025-09-27 02:17 UTC

Not for long. Chinese researchers have just developed an AI search called Spike Brain that tries to mimic the human brain. It runs 90% more efficiently than current models. It's ready for use, and like all things AI, it will be worse today than it will ever be in the future.

u/CodFull2902 • 1 points • 2025-09-27 01:14 UTC

For now, its not exactly clear whether future models will still require as much computing power and electricity. Its pretty likely we will find a way to optimize this

u/borringman • 7 points • 2025-09-27 00:26 UTC

The Achilles heel is that it scales exponentially in a very BAD way. These are placeholder numbers so don't quote them but the basic idea is that you can improve the accuracy of your output by, say, 3% if you double your costs. So 6% is quadruple, 9% octuple, until you're bleeding the entire country dry but still nowhere near an acceptable success rate. This is the exact opposite of how you want economies of scale to work.

The "geniuses" running the show are too stupid to understand math I learned in high school.

u/pastfuturologycheck • 10 points • 2025-09-27 00:23 UTC

No, the death of Moore's law is. Making bigger, more expensive and power hungry datacenters is the futile attempt to counterbalance the fact that progress in computing power is stagnating.

A single GB200 that costs \$60,000 has 62Tflops of FP64 performance and 124Tflops of FP32 performance. This is roughly equivalent to the fastest supercomputer from [2004] (<https://top500.org/lists/top500/2004/11>).

A PS2 back in 2000 cost \$300 and had 6Gflops of FP32 performance. That was roughly equivalent to a [Cray Y-MP from 1988](https://en.wikipedia.org/wiki/Cray_Y-MP).

An XBOX back in 2001 cost \$300 and had 20Gflops of FP32 performance. That was roughly equivalent to a [Cray XMS from 1990](https://en.wikipedia.org/wiki/Cray_XMS).

u/cdca • 21 points • 2025-09-26 23:46 UTC

The Achilles Heel of AI is that it doesn't fucking work.

u/iBN3qk • 1 points • 2025-09-27 02:05 UTC

Isn't it just ML for the masses?

u/syzorr34 • 2 points • 2025-09-27 14:09 UTC

No.

ML is often considered to be a subset of "statistical" models (I put statistics in quotes there because even though they use the same mathematics as statistics, they are nowhere near as robust as actual statistical models) where the goal is for determining

correlation by throwing away any hope of establishing causation.

LLMs are that taken to an even greater extreme where it's nothing but correlations, but also with zero understanding.

u/SwimmingBarracuda182 • 2 points • 2025-09-26 23:05 UTC

It's more so that it's known to hallucinate at any basic level of complexity. In almost all industries, this is a non-starter.

u/OkCar7264 • 4 points • 2025-09-26 22:54 UTC

I mean kinda. I think when they start charging what it actually costs a lot of people are going to feel like they can just watch the YouTube video themselves instead of a 400/mo subscription.